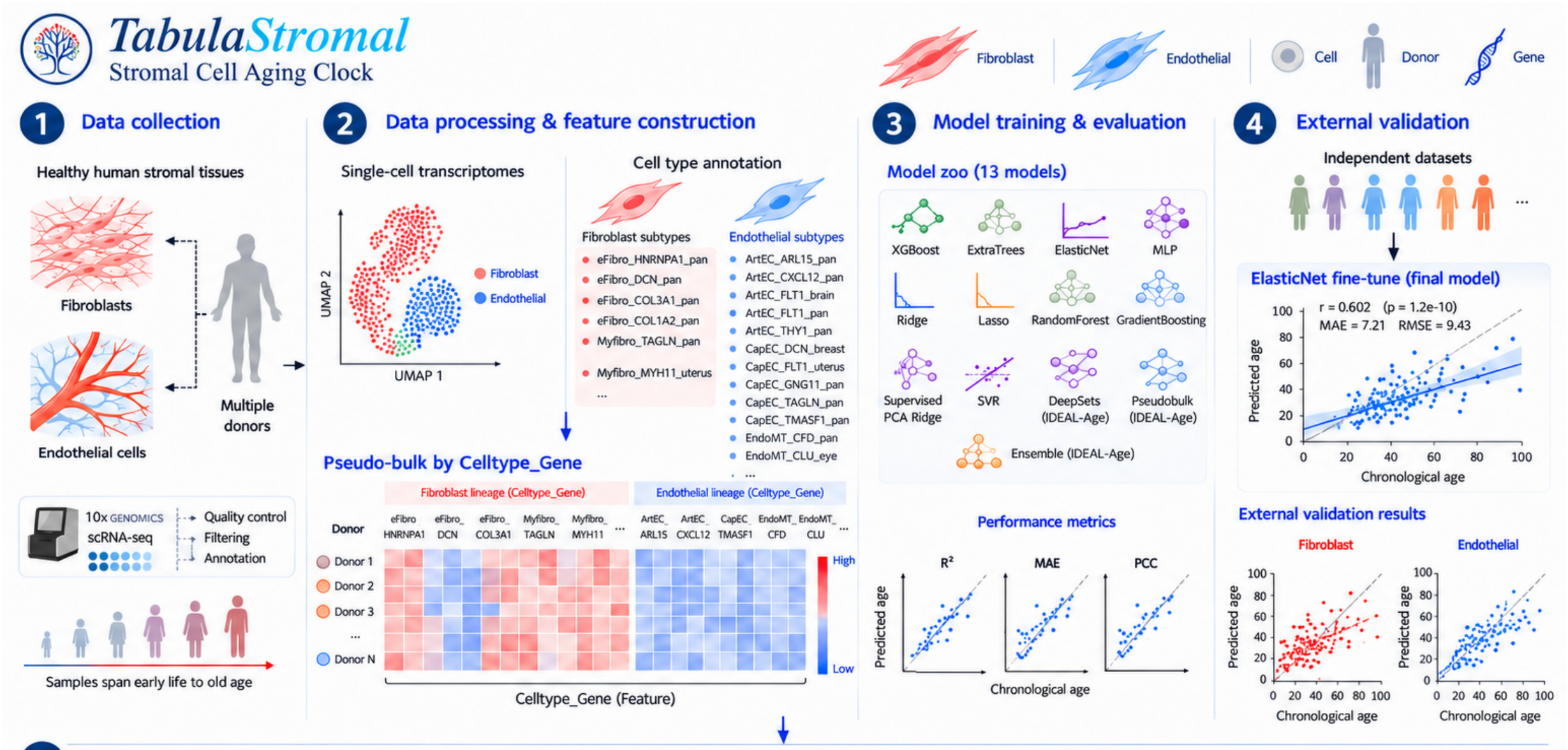
A StromalClock workflow



B R^2

	Fibroblast			Endothelial		
	Train	Valid	External	Train	Valid	External
XGBoost	0.982	0.783	-0.824	0.936	0.697	-0.388
ExtraTrees	0.976	0.760	-0.634	0.924	0.759	-0.839
ElasticNet	0.930	0.694	0.410	0.877	0.619	0.218
MLP	0.966	0.659	-0.152	0.937	0.676	0.048
Ridge	0.984	0.409	-0.268	0.942	0.517	-0.697
Lasso	0.849	0.664	0.249	0.834	0.595	0.100
RandomForest	0.913	0.710	-1.232	0.848	0.694	-0.677
Grad. boost	0.897	0.676	-0.351	0.877	0.688	-0.001
SPCA–Ridge	0.602	0.505	-2.120	0.607	0.461	-1.601
SVR	0.659	0.409	-2.629	0.602	0.455	-2.110
DeepSets	0.977	0.696	-1.117	0.934	0.718	-0.523
Pseudobulk	0.984	0.621	-1.693	0.938	0.648	-1.092
Ensemble	0.983	0.723	-1.418	0.939	0.730	-0.602

-2.63 -1.00 0.00 0.98

Negative (blue) · 0 · Positive (red)

C MAE (years)

	Fibroblast			Endothelial		
	Train	Valid	External	Train	Valid	External
XGBoost	1.69	10.57	18.12	3.42	12.76	17.06
ExtraTrees	2.31	11.13	16.20	3.89	11.06	17.06
ElasticNet	4.84	13.33	8.95	7.52	14.75	8.77
MLP	3.23	12.05	12.75	3.38	11.90	10.21
Ridge	0.96	17.11	12.18	2.38	14.67	14.93
Lasso	8.90	14.58	9.85	9.55	15.48	10.54
RandomForest	6.21	12.64	20.12	8.05	12.61	18.58
Grad. boost	6.00	13.47	15.76	6.69	13.36	13.21
SPCA–Ridge	15.04	18.15	26.36	14.88	18.33	24.43
SVR	11.36	16.33	27.18	11.77	15.19	26.28
DeepSets	2.83	11.67	17.63	4.15	10.50	16.98
Pseudobulk	1.67	15.03	22.32	3.48	13.77	19.72
Ensemble	1.71	12.42	19.75	3.47	11.39	17.86

0.96 9.70 18.44 27.18

Darker = higher MAE

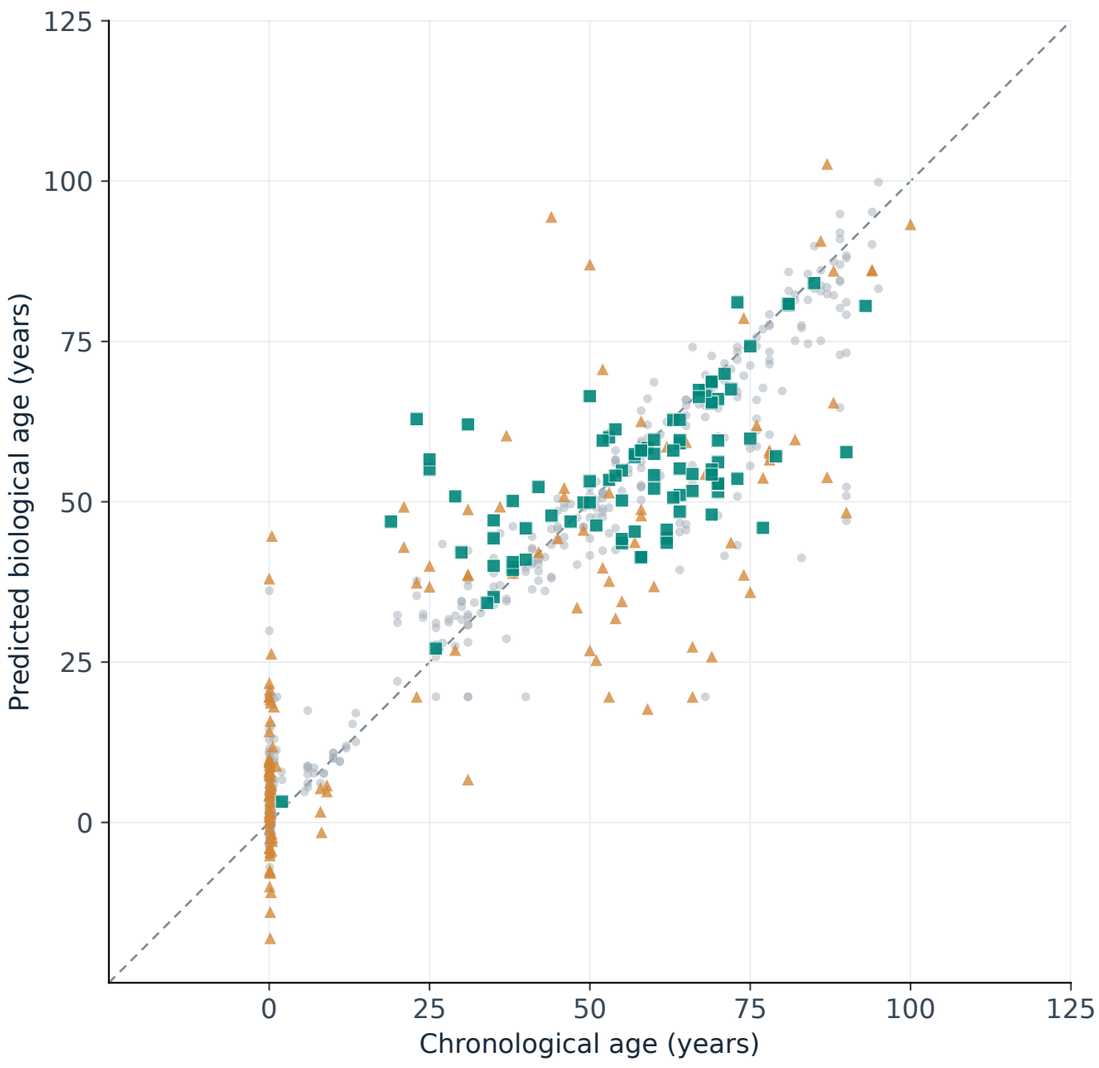
D MAD (years)

	Fibroblast			Endothelial		
	Train	Valid	External	Train	Valid	External
XGBoost	28.49	28.89	13.21	28.30	28.31	13.37
ExtraTrees	28.49	28.89	13.21	28.30	28.31	13.37
ElasticNet	28.49	28.89	13.21	28.30	28.31	13.37
MLP	28.49	28.89	13.21	28.30	28.31	13.37
Ridge	28.49	28.89	13.21	28.30	28.31	13.37
Lasso	28.49	28.89	13.21	28.30	28.31	13.37
RandomForest	28.49	28.89	13.21	28.30	28.31	13.37
Grad. boost	28.49	28.89	13.21	28.30	28.31	13.37
SPCA–Ridge	28.49	28.89	13.21	28.30	28.31	13.37
SVR	28.49	28.89	13.21	28.30	28.31	13.37
DeepSets	28.49	28.89	13.21	28.30	28.31	13.37
Pseudobulk	28.49	28.89	13.21	28.30	28.31	13.37
Ensemble	28.49	28.89	13.21	28.30	28.31	13.37

13.21 18.44 23.66 28.89

Darker = higher MAD

E Fibroblast



F Endothelial

